Sleep Tracker

Full-Stack Web Application

Angular 17 | Django 5 | DRF | JWT Auth | SQLite

Track your sleep · Understand your habits · Improve your rest

Web Development Project · 2024

What Was Built

User Authentication

Register, login, logout with JWT tokens. Passwords hashed securely.

Add Sleep Records

Log sleep time, wake time, quality (1–5) and optional notes.

View All Records

Full table of all past sleep sessions with date, duration, quality badge.

Edit & Delete

Update any record via modal form. Delete with one click.

Statistics Page

Aggregated stats: averages, longest/shortest sleep, quality distribution bar chart.

Profile & Goals

User profile editor and personal sleep goal setter (target hours & quality).

Backend Requirements · Django + DRF

Models & Database

SleepRecord	Sleep session — date, times, quality, notes, duration
UserProfile	Extended user data — bio, timezone
SleepCategory	Tags for records — FK → User
SleepGoal	Target hours & quality — FK → User
Custom Manager	SleepRecordManager: for_user(), this_week(), good_quality()

Serializers

serializers.Serializer	LoginSerializer, SleepStatsSerializer
ModelSerializer	SleepRecordSerializer, UserSerializer, CategorySerializer, GoalSerializer

Views

FBV	login_view — validates credentials, returns JWT
FBV	logout_view — blacklists refresh token
FBV	register_view — creates user + profile + goal
FBV	sleep_stats_view — aggregated statistics
CBV	SleepRecordListCreateView — GET list / POST create
CBV	SleepRecordDetailView — GET / PUT / PATCH / DELETE
CBV	SleepCategoryListCreateView — categories CRUD
CBV	UserProfileView — GET & PATCH profile + goal

✓ JWT Auth
login + logout endpoints

✓ CORS configured
Allows localhost:4200

Frontend Requirements · Angular 17

Data & Services

✓ Angular Service

SleepService uses HttpClient for ALL API calls — records, stats, categories, profile

✓ TypeScript Models

Interfaces matching every Django response shape — SleepRecord, SleepStats, UserProfile…

✓ Error Handling

handleError() in SleepService — friendly messages for 0, 401, 403, 404, 500

Forms & Events

✓ 4+ [(ngModel)]

sleep_time, wake_time, quality, notes on Dashboard · username, password on Login · goal fields on Profile

✓ 4+ (click) events

Add Record · Edit (save) · Delete · Login · Register · Save Profile · Save Goal · Refresh

✓ FormsModule

Imported in app.config.ts — enables all two-way data binding

Architecture

✓ 6 Named Routes

/login · /register · /dashboard · /records · /statistics · /profile

✓ @for + @if

@for on all record tables · @if for loading, empty, error states throughout all pages

✓ JWT Interceptor

AuthInterceptor auto-attaches Bearer token · AuthGuard protects all routes · logout blacklists token

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✓ 4 Models · ✓ 2+ FBVs + 2+ CBVs · ✓ 2 Serializer + 2 ModelSerializer · ✓ Full CRUD · ✓ JWT Auth · ✓ CORS · ✓ Postman Collection · ✓ 4 ngModel + 4 click events · ✓ @for + @if · ✓ 3+ Routes